

Corporate Finance

Lecture 4: CAPM

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Plan

- ▶ Recap from lecture 3
- ▶ Ask what return a stock must offer to be worth adding to a portfolio: the required return and the beta
- ▶ Meet the CAPM: when every investor holds the tangency portfolio, that portfolio is the market itself
- ▶ Use the CAPM to determine the risk premium of any stock

Mostly based on the BDM textbook Chap. 11.6-8.

Recap lecture 3

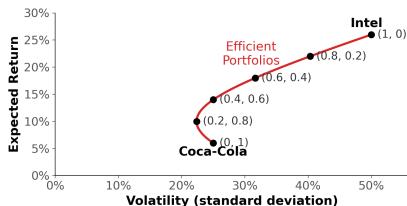
Recap of Lecture 3

Let's do a recap:

- ▶ Combining two stocks: many portfolios, many risk-return pairs
- ▶ Efficient portfolios and the efficient frontier
- ▶ The shape of the frontier depends on the correlation
- ▶ The minimum-variance portfolio
- ▶ Short sales extend the frontier
- ▶ From two stocks to N : the frontier moves left
- ▶ Adding the risk-free asset
- ▶ Borrowing and leverage
- ▶ The Sharpe ratio, and how it picks the efficient frontier
- ▶ The super-efficient (tangency) portfolio
- ▶ Two types of risk

Recap: combining two stocks

- ▶ Given two stocks, we choose the **weights** x_1 and x_2 , with $x_1 + x_2 = 1$. Each choice is one portfolio, with its own expected return and its own risk.
- ▶ Plotted as points, risk across and return up, all the mixes of the two stocks trace a **curve** (Intel and Coca-Cola, zero correlation):



Recap: efficient portfolios and the frontier

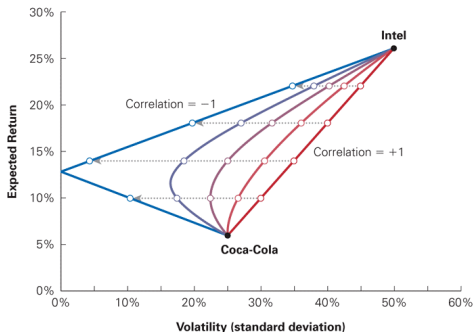
A portfolio is **efficient** when, for its level of risk, no other portfolio gives a higher return; equivalently, for its level of return, no other portfolio has a lower risk.

The **efficient frontier** is the set of all efficient portfolios: the upper part of the curve.

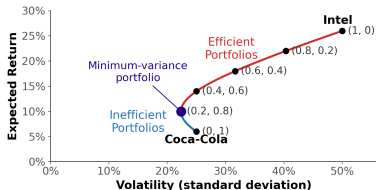
- ▶ Example: $(0, 1)$ and $(0.4, 0.6)$ both have 25% volatility, but the second returns 14% against 6%. Holding only Coca-Cola is dominated; the lower part of the curve is inefficient.

Recap: the shape depends on correlation

- ▶ **Correlation +1:** no diversification. The curve is a straight line between the two stocks.
- ▶ **Lower correlation:** the curve bends to the left, the same returns at lower risk.
- ▶ **Correlation -1:** the risks can cancel completely; with the right weights the portfolio has zero risk.



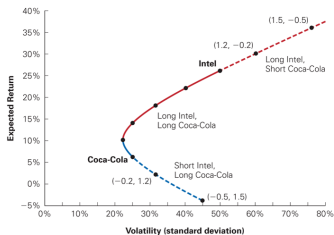
Recap: the minimum-variance portfolio



- ▶ The mix with the **lowest risk** of all: the “nose” of the curve, where the efficient set begins. Here (0.2, 0.8), with 22.4% volatility, below either stock alone.
- ▶ Its weights depend only on the volatilities and the correlation, **not on the expected returns**.

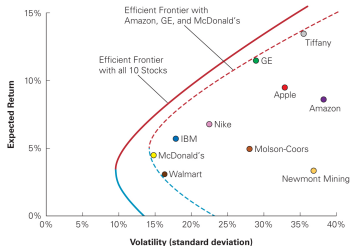
Recap: short sales extend the frontier

- ▶ A **short sale** is a negative weight: sell a stock you borrowed, and invest more than your own money in the other one.
- ▶ Shorting assets allows to extend the efficient frontier further.



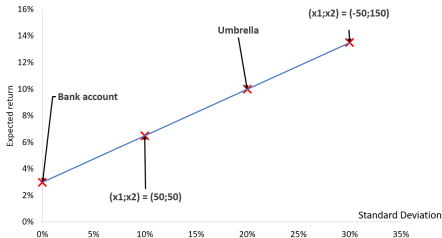
Recap: from two stocks to N

- ▶ A third stock turns the curve into a **region** of possible portfolios. The efficient frontier is its **upper-left edge**.
- ▶ Every stock we add, even a poor one on its own, moves the frontier further left: more diversification. The best frontier uses all available investments.



Recap: adding the risk-free asset

- ▶ The risk-free rate r_f is known in advance: zero volatility, zero correlation with every stock.
- ▶ A fraction x in a risky portfolio and $1 - x$ at r_f gives $E[R_p] = r_f + x(E[R_U] - r_f)$ and $\sigma_p = x\sigma_U$: both grow with x , so every mix is on a **straight line** from the risk-free point. Umbrella (10%, 20%) and a 3% deposit, half and half: 6.5% and 10%.



Recap: borrowing and leverage

- ▶ **Leverage** is investing with borrowed money. In the formulas it is a **negative weight on the risk-free asset**, $x_f < 0$, with more than 100% in the risky one.
- ▶ Borrow 50 at 3% and put 150 in Umbrella: $x_U = 1.5$, $x_f = -0.5$. Return 13.5% instead of 10%, volatility 30% instead of 20%: leverage scales the risk premium and the risk by the **same factor**.
- ▶ It takes you beyond the risky portfolio on the same straight line. Gains and losses are both magnified, so **never judge a return without its risk**.

Recap: the Sharpe ratio

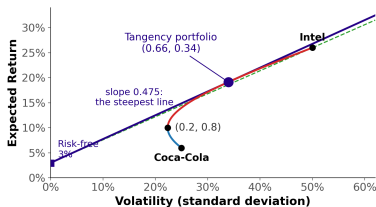
- ▶ The slope of the line from the risk-free point is the **Sharpe ratio**: the extra return you get per unit of risk.

$$\text{Sharpe ratio} = \frac{E[R] - r_f}{\sigma}$$

Portfolio	Expected return	Volatility	Sharpe ratio
Coca-Cola (0, 1)	6%	25%	0.12
(0.2, 0.8)	10%	22.4%	0.31
(0.5, 0.5)	16%	28%	0.46
Intel (1, 0)	26%	50%	0.46

- ▶ A steeper line pays more at every level of risk, whoever you are. With leverage any return is possible, so **compare Sharpe ratios, not returns**.

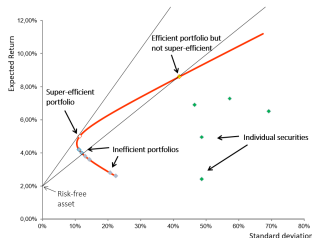
Recap: the Sharpe ratio and the efficient frontier



- The steepest line from the risk-free point is the one **tangent** to the curve. Once a risk-free asset exists, the efficient frontier **is that straight line**:

$$E(R_p) = r_f + \underbrace{\frac{E(R_T) - r_f}{\sigma_T}}_{\text{Sharpe ratio of } T} \times \sigma_p \quad \text{here } 3\% + 0.475 \times \sigma_p$$

Recap: the super-efficient portfolio

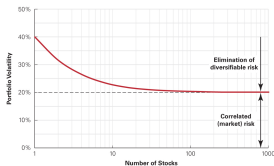


- ▶ The tangency point T is the risky portfolio with the **highest Sharpe ratio**: the tangency, or **super-efficient**, portfolio.
- ▶ **Everyone holds it.** Risk aversion only sets where you sit on the line: cautious investors lend part of their money at r_f , bold ones borrow and put more than their own money in T (two-fund separation).

Recap: two types of risk

Systematic risk: market-wide (economy, interest rates, inflation). It hits every stock at once, so diversification **cannot** remove it.

Idiosyncratic risk: specific to one firm (its products, its management). It averages out across many stocks, so diversification **does** remove it.



Volatility falls as stocks are added and flattens at the systematic level.

Beginning of lecture 4 content

Lecture 3 found the best portfolio of risky assets. Today we ask what that implies for the return each stock has to offer: the CAPM.

Introduction

- ▶ Today: how to determine the risk premium a stock must offer, with the **Capital Asset Pricing Model (CAPM)**: Tobin (1958), Treynor (1962), Sharpe (1964), Lintner (1965), Mossin (1966).
- ▶ **Why it matters.** The CAPM is the most widely used model linking risk and return. Most firms use it to measure the cost of capital: the discount rate for projects, valuations and regulated prices.



James Tobin



William Sharpe



Harry Markowitz

Outline

- A. Required returns: is a stock worth adding to a portfolio? (Chap. 11.6)
- B. From the tangency portfolio to the market portfolio (Chap. 11.7)
- C. The CAPM, and how to read it (Chap. 11.8)

A. Required returns: the answer first

- ▶ Last class we met the **Sharpe ratio**: the extra return we get for every unit of risk we take. Today we use it to decide whether an asset is **worth adding to a portfolio**.
- ▶ You hold a portfolio P . Someone offers you a stock A . **Should you buy some?**

The answer, in one sentence. A stock is worth adding only if its expected return pays for the risk it adds to *your* portfolio. And the risk it adds depends on how it **moves with what you already hold**, not on how risky it is on its own.

- ▶ The rest of this section unpacks that sentence in three steps:
 1. measure the risk A adds to P ;
 2. measure what that extra risk is worth;
 3. compare with what A actually pays.

Step 1: the risk a stock adds

- ▶ Add a small amount δ of A to P , financed by borrowing at r_f . The extra return is easy: $\delta(E(R_A) - r_f)$.
- ▶ The extra risk is not $\delta\sigma_A$. For a small δ the volatility of the portfolio rises by about

$$\Delta\sigma \approx \delta \times \sigma_A \times \rho_{A,P}$$

δ : the fraction of the portfolio put into A ; σ_A : the volatility of A on its own; $\rho_{A,P}$: the correlation between A and the portfolio P .

- ▶ The **correlation** is in the formula: a stock that moves with your portfolio adds its full risk, an uncorrelated one almost none, a negatively correlated one **reduces** risk. The same diversification story as Bore Industries.
- ▶ Where the formula comes from: the appendix at the end of the deck.

Step 2: what is that extra risk worth?

- ▶ You could take the same extra risk another way: **leverage**, borrowing to buy more of P .
- ▶ Leverage on P pays the Sharpe ratio of P for each unit of risk (Lecture 3). So extra risk $\Delta\sigma$ taken through P would earn

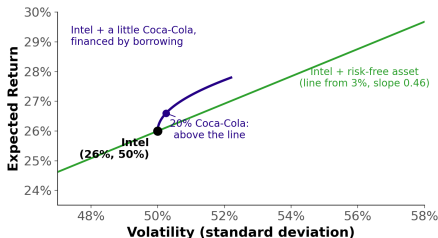
$$\Delta\sigma \times \frac{E(R_P) - r_f}{\sigma_P}$$

$\Delta\sigma$: the extra risk from step 1; $E(R_P) - r_f$: the risk premium of P ; σ_P : its volatility. The fraction is P 's Sharpe ratio.

- ▶ That is the **benchmark**. A is worth adding when its extra return (step 1, left) is at least what the same risk earns through P (right):

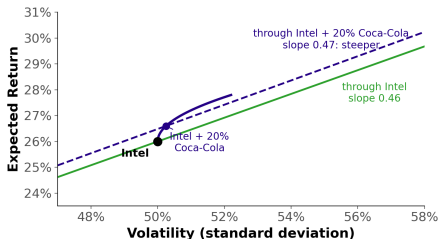
$$\delta(E(R_A) - r_f) \geq \Delta\sigma \times (E(R_P) - r_f) / \sigma_P$$

In a picture: Intel and Coca-Cola



- ▶ **Setting.** Your portfolio P is Intel (26%, 50%); the stock A on offer is Coca-Cola (6%, 25%), uncorrelated with Intel; the risk-free rate is 3%. The chart zooms in around Intel.
- ▶ **Green line:** Intel mixed with the risk-free asset, the line from Lecture 3 (slope = Intel's Sharpe ratio, 0.46). Beyond Intel it is leverage on Intel.

In a picture: Intel and Coca-Cola (con't)



- ▶ **Blue curve:** borrow a little at 3% and put it into Coca-Cola, on top of Intel. The dot is $\delta = 20\%$: return 26.6%, volatility 50.3%.
- ▶ **The new line:** draw the line from the risk-free rate through that dot. It is **steeper** than the line through Intel: Sharpe ratio 0.47 against 0.46. Adding a little Coca-Cola **raises the Sharpe ratio**, at every level of risk.

Step 3: from the inequality to the rule

- Start from the inequality of step 2 and put in the extra risk from step 1, $\Delta\sigma = \delta\sigma_A\rho_{A,P}$:

$$\delta(E(R_A) - r_f) \geq \delta\sigma_A\rho_{A,P} \times \frac{E(R_P) - r_f}{\sigma_P}$$

- δ appears on both sides, so the size of the purchase does not matter: **cancel it**.

$$E(R_A) - r_f \geq \sigma_A\rho_{A,P} \times \frac{E(R_P) - r_f}{\sigma_P}$$

- Move the volatility of P next to the other risk terms:

$$E(R_A) - r_f \geq \underbrace{\rho_{A,P} \frac{\sigma_A}{\sigma_P}}_{\text{a single number}} \times (E(R_P) - r_f)$$

Step 3: that single number is the beta

- ▶ **Remember the beta** from Lecture 2, the third monster: the covariance of two things divided by the variance of the one that moves. It tells you how much one moves when the other moves by 1%.
- ▶ Here the thing that moves is the portfolio P , and we ask how much A moves with it. Using $\text{Cov}(R_A, R_P) = \rho_{A,P} \sigma_A \sigma_P$:

$$\beta_{A,P} = \frac{\text{Cov}(R_A, R_P)}{\text{Var}(R_P)} = \frac{\rho_{A,P} \sigma_A \sigma_P}{\sigma_P^2} = \rho_{A,P} \frac{\sigma_A}{\sigma_P}$$

Exactly the single number from the previous slide.

- ▶ So the rule reads: buy A if $E(R_A) - r_f \geq \beta_{A,P}(E(R_P) - r_f)$. The beta is how we will use comovement from now on: it scales P 's risk premium into the premium A must offer.

Step 3: the required return

- Move r_f to the right-hand side. The rule becomes a **required return** r_A :

$$E(R_A) \geq r_A = r_f + \beta_{A,P}(E(R_P) - r_f)$$

- Read it as: **risk-free rate, plus the portfolio's risk premium scaled by how much A moves with the portfolio.**
 - $\beta_{A,P} = 1$: A must pay the same premium as P .
 - $\beta_{A,P} = 0.5$: half the premium is enough, because A adds only half as much risk.
 - $\beta_{A,P} < 0$: A reduces risk, so it is worth holding even below r_f .
- Buy A if it pays at least r_A ; if it pays less, the best move is to short it.

Example: should we add Coca-Cola to Intel?

- ▶ You hold Intel: expected return 26%, volatility 50%. Treasury bills: 3%. Coca-Cola: expected return 6%, volatility 25%, correlation with Intel 0.
- ▶ Beta of Coca-Cola with respect to Intel:
 $\beta = 0 \times 25\%/50\% = 0$.
- ▶ Required return: $r_{\text{Coca-Cola}} = 3\% + 0 \times (26\% - 3\%) = 3\%$.
- ▶ Coca-Cola pays 6%, above 3%: **yes, buy some**. It looks poor next to Intel (a quarter of the return), but being uncorrelated it adds almost no risk, so any return above the risk-free rate is worth having. Lecture 3 agreed: the tangency portfolio holds 34% of Coca-Cola.
- ▶ **The twist.** With a correlation of 0.5 instead of 0:
 $\beta = 0.5 \times 25\%/50\% = 0.25$, required return
 $3\% + 0.25 \times 23\% = 8.75\%$, above the 6% it pays: **no**. Same stock, same return, same risk; the correlation alone flips the decision.

From the required return to the CAPM

- ▶ The rule holds for **any** portfolio P : the required return of a stock is the risk-free rate plus its beta with respect to P times P 's risk premium.
- ▶ Lecture 3 showed that every investor holds the same risky portfolio, the tangency portfolio. Section B asks what that portfolio must be when everyone holds it, and finds the answer: **the market**.
- ▶ Replace P by the market portfolio and the required return becomes the **CAPM**.

B. From the tangency portfolio to the market

- ▶ Section A gave a rule that works for **any** portfolio P :

$$r_A = r_f + \beta_{A,P} (E(R_P) - r_f)$$

- ▶ Two questions decide everything that follows:
 1. **Which P does everyone hold?** Lecture 3 said: the tangency portfolio. But which portfolio is that, concretely?
 2. **What does that imply for each stock's return?**
- ▶ Where we are going, in four steps: the tangency portfolio turns out to be **the whole market**, and every stock must then pay

$$E(R_i) = r_f + \beta_i (E(R_M) - r_f) \quad \text{the CAPM}$$

Step 1: three assumptions

1. **Everyone can lend and borrow at the risk-free rate r_f .** So the efficient set is the straight line from r_f tangent to the frontier (Lecture 3).
2. **Everyone shares the same estimates** of expected returns, volatilities and correlations. So everyone computes the **same** frontier and the **same** tangency portfolio.
3. **Everyone holds an efficient portfolio.** So everyone holds that one tangency portfolio, mixed with the risk-free asset in their own proportion.

Strong assumptions. We look at what they imply first, and at their realism at the end.

Step 2: everyone's portfolio is the market

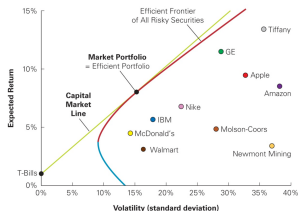
- ▶ Camille holds the tangency portfolio: EUR 10 000 of Capgemini and EUR 5 000 of Sanofi, a 2 : 1 mix.
- ▶ Tom is more cautious, so he holds less of it, but the **same mix**: his EUR 2 000 of Sanofi comes with EUR 4 000 of Capgemini.
- ▶ Now add up **every** investor. Each holds Capgemini and Sanofi 2 : 1, so the total held of Capgemini is twice the total held of Sanofi.
- ▶ So the tangency portfolio must contain every stock **in proportion to its market capitalisation**: it represents the whole universe of assets, therefore we call it: the **market portfolio** M .

In equilibrium, the tangency (super-efficient) portfolio is the market portfolio.

Step 3: the Capital Market Line

- ▶ Everyone mixes the market and the risk-free asset.
- ▶ The line that contains all the combinations of the risk free asset and the market portfolio is called the **Capital Market Line (CML)**:

$$E(R_p) = r_f + \frac{E(R_M) - r_f}{\sigma_M} \times \sigma_p$$



Step 4: every stock pays its required return

- ▶ Take any stock i and ask what happens if it did **not** pay its required return with respect to M .
- ▶ **Suppose it paid more.** By section A, adding a little of i to M would raise the Sharpe ratio. Then M would not be the best portfolio, and everyone would be buying i . Its price would rise until the extra return disappears.
- ▶ **Suppose it paid less.** Shorting a little of i would raise the Sharpe ratio. Everyone would sell, its price would fall until the gap closes.
- ▶ So in equilibrium **no stock can beat or trail its required return**: each pays exactly

$$E(R_i) = r_f + \beta_i (E(R_M) - r_f)$$

- ▶ If every asset must comply this equation, we can use it to evaluate the returns of assets. We say that it is a pricing model, because, with today's prices, if we know the expected

C. The CAPM, and how to read it

$$E(R_i) = r_f + \underbrace{\beta_i (E(R_M) - r_f)}_{\text{risk premium of stock } i} \qquad \beta_i = \frac{\text{Cov}(R_i, R_M)}{\text{Var}(R_M)}$$

- ▶ β_i is the beta of stock i **with respect to the market**: how much i moves when the market moves by 1%.
- ▶ Only the part of i 's risk that **moves with the market** is paid for. Its idiosyncratic risk earns nothing: investors can diversify it away for free, so nobody needs to be compensated for it (the two types of risk).
- ▶ $\beta_i = 1$: the market's own premium. $\beta_i = 0$: just r_f . $\beta_i < 0$: below r_f , since the stock is insurance against market falls.

Example 1: expected return of a stock

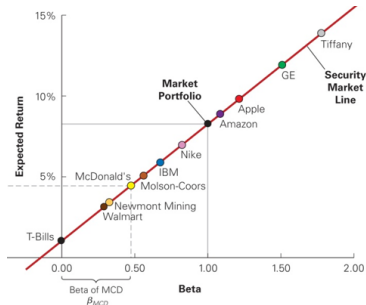
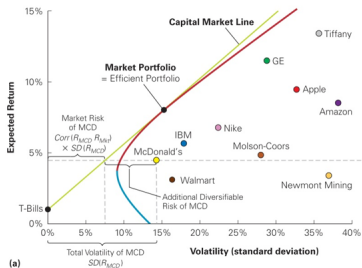
- ▶ Risk-free rate 4%. Market: expected return 10%, volatility 16%. Stock A: volatility 16%, correlation with the market 0.33.
- ▶ **Beta:** $\beta_A = 0.33 \times 16\% / 16\% = 0.33$.
- ▶ **Expected return under the CAPM:**
 $E(R_A) = 4\% + 0.33 \times (10\% - 4\%) = 6\%$.
- ▶ **The CML portfolio with the same risk.** A 16% volatility on the CML means 100% in the market: expected return 10%.
- ▶ Same volatility, 10% against 6%. Stock A carries as much risk as the market, but only a third of it is market risk; the rest is idiosyncratic and earns nothing.

Example 2: a negative beta

- ▶ Stock B has a beta of -0.3 . Risk-free rate 4% , expected market return 10% .
- ▶ **Expected return under the CAPM:**
$$E(R_B) = 4\% + (-0.3) \times 6\% = 2.2\%.$$
- ▶ Below the risk-free rate. Why would anyone hold it? Because B **risks when the market falls**: it is insurance for the rest of the portfolio, and investors pay for insurance by accepting a lower return.

Two lines: the CML and the SML

- **SML** (right, return against **beta**): the CAPM is a straight line through the risk-free asset ($\beta = 0$) and the market ($\beta = 1$). **Every** stock and portfolio sits on it.
- **CML** (left, return against **volatility**): only mixes of the market and the risk-free asset sit on it. A single stock lies **below** it, because its idiosyncratic risk adds volatility without adding return.



The beta of a portfolio

- ▶ A portfolio with weights x_i has return $R_p = x_1R_1 + x_2R_2 + \dots$. Covariance is linear, so its beta is the **weighted average of the betas**:

$$\beta_p = \frac{\text{Cov}(R_p, R_M)}{\text{Var}(R_M)} = x_1\beta_1 + x_2\beta_2 + \dots$$

- ▶ **Example.** Danone has a beta of 0.5, Ubisoft 1.8. Risk-free rate 4%, expected market return 10%. An equally weighted portfolio:

$$\beta_p = 0.5 \times 0.5 + 0.5 \times 1.8 = 1.15$$

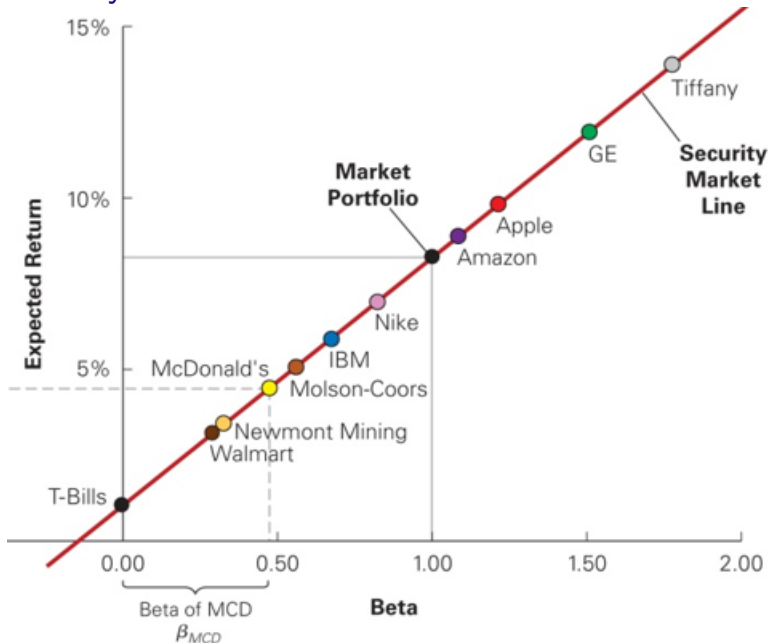
$$E(R_p) = 4\% + 1.15 \times 6\% = 10.9\%$$

Summary: the CAPM formula

$$E(R_i) = r_f + \beta_i (E(R_M) - r_f)$$

- ▶ $E(R_i)$: the **expected return of stock i** , which in equilibrium is also the return investors require to hold it.
- ▶ r_f : the **risk-free rate**, what you earn with no risk at all.
- ▶ $E(R_M) - r_f$: the **market risk premium**, the extra return the market portfolio pays above the risk-free rate.
- ▶ $\beta_i = \text{Cov}(R_i, R_M) / \text{Var}(R_M)$: the **beta** of stock i , how much it moves when the market moves by 1%. It scales the market premium into the premium of stock i .
- ▶ In words: **the return of a stock is the risk-free rate plus its share of the market premium, and that share is its beta.**

The Security Market Line



Summary: what to remember from the CAPM

- ▶ **The market portfolio is the only efficient portfolio of risky assets.** The best you can do at any level of risk is to mix it with the risk-free asset: the Capital Market Line.
- ▶ **Only market risk is paid.** A stock's premium depends on its beta, its comovement with the market; idiosyncratic risk earns nothing because it can be diversified away. This is the Security Market Line.
- ▶ **The assumptions are strong:** everyone shares the same estimates, and everyone can borrow freely at r_f . Neither is true in practice.
- ▶ **Yet the CAPM is the standard tool** for the expected return of a stock and the cost of capital of a project, as we saw in the introduction.

Appendix: portfolio volatility approximation

In Section 1 of the slides, I claim that the approximation below holds:

$$\begin{aligned}\Delta\sigma &= \sqrt{\sigma_P^2 + \delta^2\sigma_A^2 + 2\delta\sigma_P\sigma_A\rho_{A,P}} - \sigma_P \\ &\approx \delta \frac{\sigma_P\sigma_A\rho_{A,P}}{\sigma_P}.\end{aligned}$$

To prove this result, define the function f as

$$f(\delta) := \sqrt{\sigma_P^2 + \delta^2\sigma_A^2 + 2\delta\sigma_P\sigma_A\rho_{A,P}}.$$

From Taylor's theorem, it follows that for δ close to zero, we have

$$f(\delta) \approx f(0) + f'(0)\delta.$$

Appendix: portfolio volatility approximation (con't)

Moreover, it is easy to check that

$$f'(\delta) = \frac{\delta\sigma_A^2 + \sigma_P\sigma_A\rho_{A,P}}{\sqrt{\sigma_P^2 + \delta^2\sigma_A^2 + 2\delta\sigma_P\sigma_A\rho_{A,P}}},$$

and, at $\delta = 0$, f and its first derivative simplify to $f(0) = \sigma_P$ and $f'(0) = \frac{\sigma_P\sigma_A\rho_{A,P}}{\sigma_P}$. Thus, Taylor's theorem implies that

$$\sqrt{\sigma_P^2 + \delta^2\sigma_A^2 + 2\delta\sigma_P\sigma_A\rho_{A,P}} \approx \sigma_P + \frac{\sigma_P\sigma_A\rho_{A,P}}{\sigma_P}\delta.$$

The approximation on the previous slide then follows almost immediately.